

Question ID: 4edecdba

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Easy

Question

$8x + 11y = 170$

The equation gives the possible combinations of the number of **2009** premium grade Log Cabin Pennies, x , and the number of **1996** select grade Lincoln Pennies, y , in a collection that is worth a total of **\$170**. If there are **6 1996** select grade Lincoln Pennies in the collection, how many **2009** premium grade Log Cabin Pennies are in the collection?

Correct Answer: 13

Rationale

The correct answer is **13**. It's given that the equation $8x + 11y = 170$ gives the possible combinations of the number of **2009** premium grade Log Cabin Pennies, x , and the number of **1996** select grade Lincoln Pennies, y , in a collection that is worth a total of **\$170**. It's also given that there are **6 1996** select grade Lincoln Pennies in the collection, and thus $y = 6$. Substituting **6** for y in the given equation yields $8x + 11(6) = 170$, or $8x + 66 = 170$. Subtracting **66** from both sides of this equation yields $8x = 104$. Dividing both sides of this equation by **8** yields $x = 13$. Therefore, there are **13 2009** premium grade Log Cabin Pennies in the collection.